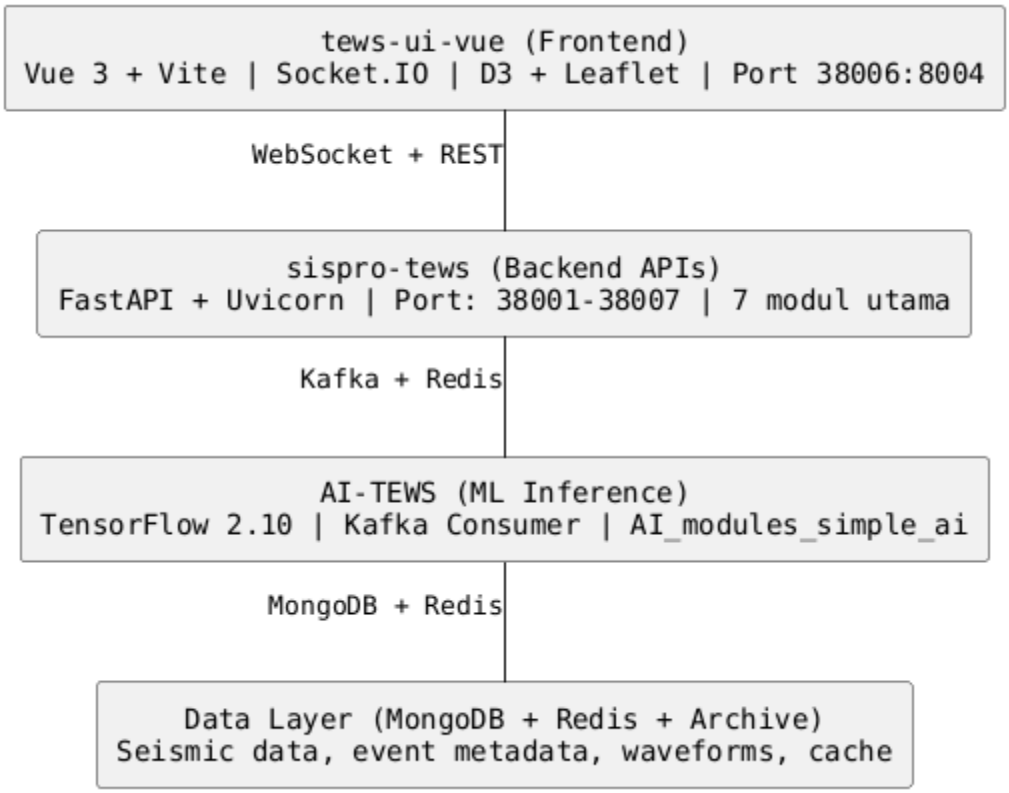
ARSITEKTUR & STRUKTUR CODEBASE

1.1 Ringkasan Komponen Utama

Sistem GEMPA AI-TEWS terdiri dari 4 komponen utama:



1.2 Rincian Setiap Komponen

A. tews-ui-vue (Frontend Web Application)

Lokasi: `./tews-ui-vue/`
Teknologi: Vue 3, TypeScript, Vite, TailwindCSS + DaisyUI
Port: 38006 (eksternal) → 8004 (internal, nginx)

Fungsi:

- Visualisasi real-time gempa dan data seismik
- Dashboard monitoring station, event, phase detection
- Integrasi peta (Leaflet, D3 untuk grafik)
- Koneksi WebSocket ke controller_module untuk live data
- REST API calls ke backend services

Dependencies Utama:

- **Vue** 3.4.31**, Vue Router** 4.4.0
- **Pinia** 2.1.7 **(state management)**
- **Socket.IO-Client** 4.7.5 **(real-time)**
- **Axios** 1.7.2 **(HTTP client)**
- **D3** 7.9.0 **(data visualization)**
- **Leaflet** 1.9.4 **(mapping)**
- **TanStack Vue Query** 5.51.1 **(data fetching)**
- **date-fns** 3.6.0**, lodash** 4.17.21

Build Process:

```
npm run build  # Output: dist/ folder
# nginx serves /app/dist ke /usr/share/nginx/html
```

Nginx Config (`tews-ui-vue/nginx.conf`):

- Listens on port 8004
- SPA fallback: semua route → index.html
- Gzip compression enabled
- Worker processes: 4

B. sispro-tews (Backend - Seismic Data Processing)

Lokasi: `./sispro-tews/`
Teknologi: Python 3, FastAPI, Uvicorn, ObsPy, MongoDB driver
Network: External Docker network `sispro-tews_req`
Port Mapping:

Modul	Port Internal	Port Eksternal	Fungsi
api_incoming_module	8007	38007	Public API untuk incoming events
record_stream_module	8001	38001	Streaming seismic records
fdsn_api_module	8002	38002	FDSN-WS API endpoint
controller_module	8003	38003	Main controller API + orchestration
websocket_waveform	8004	38004	WebSocket waveform data stream
websocket_general	8005	38005	WebSocket general updates
archiving_module	-	-	Background archiving service (no public port)
fdsn_module	-	-	FDSN scheduler (background)
seedlink_module	-	-	SeedLink protocol consumer (background)

Modul Detail:

1. api_incoming_module

- Entry point untuk data eksternal (HTTP POST)
- FastAPI app di `api_incoming_module.py`
- Wrapper: `app_incoming_module.py` → uvicorn server (32 workers, port 8007)
- Menerima: earthquake data, station info, picks, associations

2. record_stream_module

- Streaming real-time seismic waveforms
- Port 8001, volume mount ke archive location
- Consumer data dari seedlink atau external sources

3. controller_module

- Orchesterasi seluruh sistem
- Main controller API (FastAPI, port 8003)
- WebSocket untuk client connections
- 3 variants:
  - `controller_web_socket_redis.py` - Main WebSocket handler (backup & replicas)
  - `controller_web_socket.py` - Alt implementation
  - `controller_api_app.py` - API endpoint wrapper

3a. websocket_general (Dockerfile_general)

- Port 8005
- Broadcast general updates (station status, etc.)

3b. websocket_waveform (Dockerfile_waveform)

- Port 8004
- Dedicated waveform streaming (high-frequency data)
- Nginx proxy (`sispro-tews/nginx/nginx.conf`) routes WebSocket ↔ waveform service

4. fdsn_module + fdsn_api_module

- FDSN-WS (Federated Digital Seismic Network) compliant
- `fdsn_api.py` - FDSN REST API
- `fdsn_scheduler.py` - Background scheduler untuk query remote stations
- Volume mount: archive location (read seismic data)
- Port 8002 public, scheduler runs internal

5. archiving_module

- Background daemon, no public port
- Replicates: 7 instances (load-balanced archiving)
- Memory limit: 20 GB per instance
- Fungsi: persist seismic data to MongoDB/disk, cleanup old data
- Volume: `${archive_location}` (seismic archive directory)
- Docker socket mount: orchestrate sibling containers if needed

6. seedlink_module

- SeedLink protocol consumer
- Replicates: 5 instances
- Memory limit: 5 GB per instance
- Connects to remote SeedLink servers, streams waveform data
- Docker socket mount: dynamic container orchestration

C. AI-TEWS (Artificial Intelligence Modules)

Lokasi: `./AI-TEWS/`
Teknologi: Python 3, TensorFlow 2.10, Kafka, Redis, MongoDB
Versi Aktif: `AI_modules_simple_ai`

Struktur Directory:

```
AI-TEWS/
├── AI_modules_alpha/          (experimental, v1)
├── AI_modules_beta/           (experimental, v2)
├── AI_modules_neoalpha/       (experimental, v3)
├── AI_modules_simple_ai/      (RECOMMENDED - production ready)
├── AI_modules_template/       (development template)
├── requirements/              (shared dependencies: Kafka, Redis, MongoDB se
tup)
├── archive/                   (waveform data cache)
└── dump.rdb                   (Redis persistence snapshot)
```

AI_modules_simple_ai Components:

1. p-pick_module (P-wave Phase Picker)

- **Teknologi:** TensorFlow 2.10 (neural network model)
- **Memory:** 5 GB per instance, 5 replicas
- **Service** (`p-pick_module/service/`):
    - Flask/Gunicorn server (port 8000 internal)

    - Model inference engine
    - Requirements: tensorflow==2.10, keras, numpy, scipy
- **Nginx Load Balancer** (`p-pick_module/nginx/`):
    - Distributes requests to 5 service replicas
    - Nginx config: load balancing upstream
- **Consumer** (`p-pick_module/consumer/`):
    - Kafka consumer: listens to "event" topic
    - Posts waveform data to p-pick service via HTTP
    - Publishes results back to Kafka
- **Ports:**
    - Service: 8000 (internal, no external exposure - behind nginx)
    - Nginx: configurable via `${nginx_port}` in .env (default: 80)

2. association_module

- Event-phase association algorithm
- PyOcto/custom algorithm
- Consumes picks from Kafka, publishes associated events

3. phase-arrival_module

- Phase arrival time prediction
- Velocity models, travel time calculations

4. locmag_module

- Location and magnitude estimation
- Hypocenter determination using triangulation
- Magnitude calculation from amplitude

Requirements File (`AI-TEWS/AI_modules_simple_ai/p-pick_module/service/requirements.txt`):

```
kafka_python==2.0.2
obspy==1.4.0
pymongo==4.6.2
python-dotenv==1.0.1
redis==3.5.3
tensorflow==2.10
pandas==1.4.4
tqdm==4.66.4
flask
gunicorn
```

Docker Compose (`AI-TEWS/AI_modules_simple_ai/docker-compose.yml`):

- Services: p-pick_service, p-pick_nginx, p-pick_consumer, association_module, phase-arrival_module, locmag_module
- Environment: CUDA_VISIBLE_DEVICES=-1 (CPU-only; set to 0 for GPU if available)
- Dependencies: .env file for Kafka, Redis hosts

D. Infrastructure & Data Layer

1. MongoDB

- Container: Managed via sispro-tews network
- Port: 27017 (internal), 27018 (external, per memory)
- Data: `./data/` folder (WiredTiger database files)
- Collections:
  - `stations` - Network/station metadata
  - `events` - Earthquake events (origin, magnitude, etc.)
  - `picks` - Phase picks (P, S arrivals)
  - `associations` - Event-phase associations
  - `waveforms` - Seismic waveform snippets
  - `metadata` - Catalog metadata

2. Redis

- Role: Cache layer, session store, Kafka offset tracking
- Port: 6379 (internal)
- Persistence: `AI-TEWS/dump.rdb` (RDB snapshot)
- Data structure: Strings (cache), Hashes (state), Lists (queues)

3. Kafka

- Message broker for inter-service communication
- Topics: `event`, `pick`, `association`, `waveform`
- Ports: configurable (internal)
- Used by: AI modules (consumers), archiving (producers), controller (bidirectional)

4. Archive Storage

- Volume mount: `${archive_location}` (local filesystem)
- Stores: Raw seismic waveforms (miniSEED format)
- Size: ~1-2 TB typical (depends on station count & retention)

5. Data Files

- `./data.zip` - MongoDB backup (compressed, ~1.8 MB)

- `./data/` - Expanded MongoDB data (WiredTiger storage engine)
- `AI-TEWS/dump.rdb` - Redis persistence (RDB format, ~1 KB)

1.3 Data Flow (Alur Data Sistem)

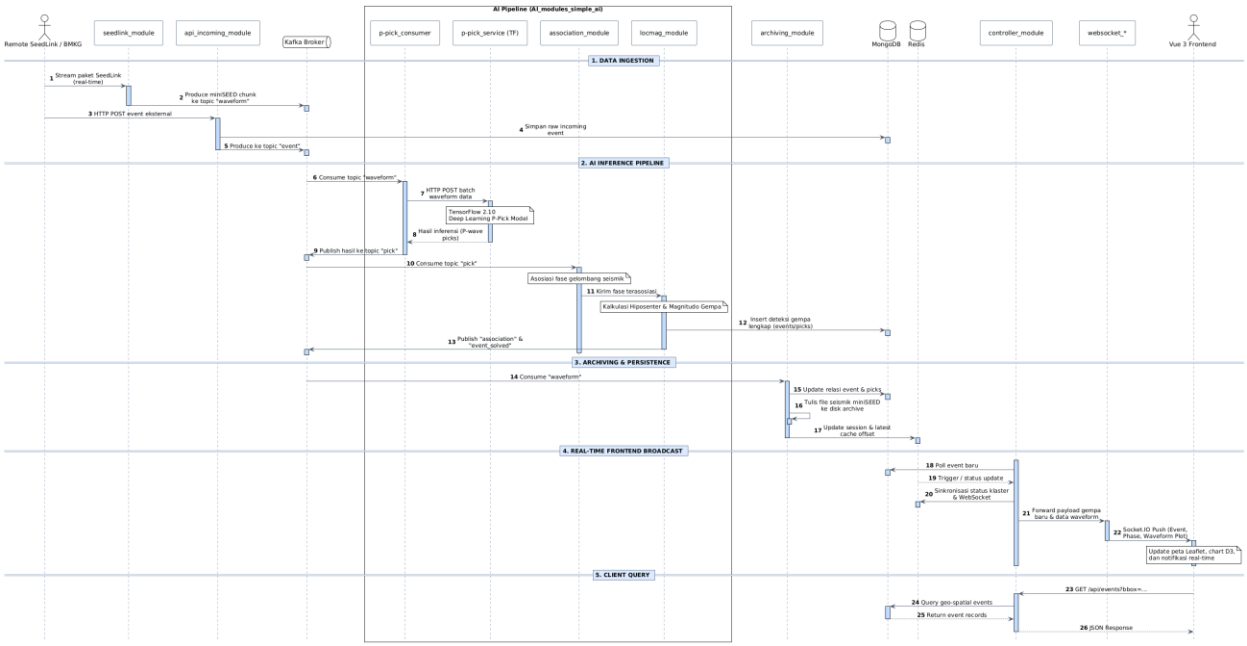
```
1. INCOMING DATA STREAM
   ├─ External SeedLink Servers
   │  └─→ seedlink_module (5x) —→ Kafka topic: "waveform"
   ├─ HTTP POST (api_incoming_module:38007)
   │  └─→ FastAPI —→ Validate —→ MongoDB + Kafka: "event"
   └─ FDSN-WS Query (client calls fdsn_api_module:38002)
      └─→ Query MongoDB, return FDSN-format response

2. AI INFERENCE PIPELINE
   ├─ P-pick Service (5x replicas)
   │  ├─ Consumer reads Kafka: "waveform"
   │  ├─ POST to p-pick_nginx (load-balanced)
   │  ├─ Model inference (TensorFlow GPU/CPU)
   │  └─ Publish Kafka: "pick"
   ├─ Association Module
   │  ├─ Consumes: "pick"
   │  └─ Publishes: "association"
   ├─ Phase-Arrival Module
   │  └─ Refines picks with travel time
   └─ LocMag Module
      └─ Calculates hypocenter, magnitude

3. ARCHIVING & PERSISTENCE
   ├─ Archiving Module (7x) reads Kafka: "waveform"
   │  ├─ Write to MongoDB (events, picks, associations)
   │  ├─ Write to filesystem archive (miniSEED)
   │  └─ Update Redis cache
   └─ FDSN Module periodically syncs remote stations

4. REAL-TIME FRONTEND UPDATES
   ├─ Controller Module (orchestrator)
   │  ├─ Polls MongoDB for new events
   │  ├─ Aggregates system status
   │  └─ WebSocket broadcast (socket.io)
   ├─ websocket_general:38005 —→ Browser (general updates)
   ├─ websocket_waveform:38004 —→ Browser (waveform plot data)
   └─ Frontend: Vue app consumes via socket.io-client

5. CLIENT QUERIES
   ├─ HTTP GET /api/events?bbox=... (controller_module:38003)
   ├─ HTTP GET /fdsnws/event/1/query? (fdsn_api_module:38002)
   ├─ HTTP POST /api/event (api_incoming_module:38007)
   └─ WebSocket: /socket.io/ (controller + websocket modules)
```



1.4 Komunikasi Inter-Service

Source	Destination	Protocol	Data	Port
seedlink_module	Kafka	Kafka	Waveform	9092
p-pick_service	Kafka	Kafka	Picks	9092
archiving_module	MongoDB	TCP	Event metadata	27017
controller_module	Redis	TCP	Cache/state	6379
websocket_*	Frontend	WebSocket	Live data	8004, 8005
Browser	controller_module	HTTP REST	Queries	38003
Browser	fdsn_api_module	HTTP FDSN	FDSN queries	38002
External	api_incoming_module	HTTP REST	New events	38007

1.5 Network & Volumes

Docker Network: `sispro-tews_req` (external, pre-created)

- All sispro-tews services + AI services connected
- Cross-service hostname resolution: `service_name:port`

Volumes:

- `${archive_location}` - Seismic archive (host path, mounted R/W)
- `/var/run/docker.sock` - Docker socket (archiving_module, seedlink_module)
    - Purpose: Orchestrate sibling containers (scaling, cleanup)

Environment Variables (.env):

- `archive_location` - Host path for archive storage
- `archive_replica_count` - Archiving module replicas (default 7)
- `seedlink_replica_count` - SeedLink module replicas (default 5)
- `kafka_host`, `kafka_port` - Kafka broker
- `redis_host`, `redis_port` - Redis connection
- `database_host`, `database_port` - MongoDB connection
- `nginx_port` - Nginx port (AI modules)